Three tiers of annotation in a whole genome: the GIAB HG002 case study

Working document: biological story, main-text fragments, Figure 1D and a Supplementary Data section

vepyr paper

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How to read this document. Each block is tagged with its intended destination: **MAIN TEXT** (Results, Discussion, Figure 1D), **SUPPLEMENTARY** (a condensed section for `vepyr-supplementary.tex`, with the full version in Appendix A) or **NOT FOR THE PAPER** (the story for the authors and notes on the analysis). Draft on branch `paper/hg002-tiered-annotation`, not merged. Every number is generated from the analysis outputs (`analysis.py` → `results/` → `make_tex.py`); rerunning the analysis updates text, tables and figures together.

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1 The biological story

NOT FOR THE PAPER Background for the authors; the paper uses the condensed fragments of Sections 2–4.

Why HG002. HG002 (NA24385) is the son in the Ashkenazi Jewish trio of the Genome in a Bottle consortium, a Personal Genome Project participant with open consent and the most thoroughly characterised human genome [13, 12]. Three properties make it a good test bed for an annotation paper: a curated benchmark call set (4,096,123 small variants on chromosomes 1–22 after decomposition) with explicit benchmark regions; matching benchmark VCFs for both parents (HG003, HG004), which allow inheritance to be checked; and a founder population, in which global allele frequencies can be misleading. The question we ask is the classic one of MacArthur et al. [8]: *how many genes does a healthy person knock out, which of these calls are real, and what does this genome carry that a clinician would care about?* Each step of the answer needs a different kind of annotation, which is the point of the vepyr design.

1. A healthy genome carries tens of real LoF variants, and many more false ones. The core annotation finds 158 variants with a pLoF consequence (stop-gained, frameshift, canonical splice site) in a MANE Select transcript [9]. Filters in the spirit of LOFTEE [7] reduce them to 57 high-confidence pLoF variants in 52 genes, 12 of them homozygous, in line with earlier estimates of about a hundred LoF variants and about twenty homozygous LoF genes per genome [8], given that our counts are restricted to MANE transcripts and benchmark regions. The removed calls form recognisable classes, and each class comes from a different tier:

- *Tier 1 (core cache).* 55 premature stops in the last exon, which escape nonsense-mediated decay. 13 calls where the "LoF" allele is near-fixed in gnomAD: pairs of nearby frameshifts in *SON* and *SLAIN1* that restore the reading frame (GRCh38 carries a rare sequence there), and *FMO2* p.Gln472Ter, the *FMO2*2* allele that truncates FMO2 in most humans [4]—a real, but species-wide, loss.
- *Tier 2 (plugins).* 39 variants classified benign in ClinVar, including *LPL* p.Ser474Ter (S447X), a stop-gain associated with lower triglycerides [10], and the *CYP2D6*4* splice-acceptor allele, a genuine loss of function in a pharmacogene that ClinVar classifies as a drug-response allele. 3 canonical splice-site variants without SpliceAI support [6].
- *Tier 3 (ad hoc).* 24 calls in segmental duplications (olfactory receptors, *NBPF*, *GOLGA8*, *NPIPB*, *CYP2D6*) and 10 in low-mappability regions, from GIAB stratifications [5]; 4 premature stops within 50 bp of the last exon junction, from Ensembl exon coordinates.

2. Purifying selection is visible in one genome. Joining the frame with gnomAD v4.1 gene constraint (read from the gnomAD bucket by URL) shows that only 9.5% of HG002 pLoF variants fall in the 20% most LoF-intolerant genes, against 24.7% of rare synonymous variants ($p = 2.9 \times 10^{-4}$), and none of the rare high-confidence ones does. A healthy individual carries LoF variants, but in genes that tolerate them. The rare synonymous reference matters: constrained genes are longer, so a naive uniform expectation (20%) would understate the depletion.

3. One clinically classified variant: an Ashkenazi founder allele. The only ClinVar pathogenic or likely pathogenic variant with review criteria is *F11* p.Glu135Ter, the factor XI "type II" allele (Glu117Ter in legacy numbering), one of the two founder mutations that account for most factor XI deficiency in Ashkenazi Jews [2]. HG002 is heterozygous; the allele was transmitted by the father; its gnomAD frequency is 0.063% overall but 1.85% in Ashkenazi Jewish samples (29-fold). The same founder effect shows across the rare pLoF variants: 4 of the 10 that are rare in gnomAD overall exceed 1% in Ashkenazi Jewish samples, so "rare" depends on the reference population. Population-specific frequencies come from the core cache; ClinVar from a plugin cache; the parental genotypes from the parents' VCFs streamed from S3.

4. The trio resolves origins and exposes one false homozygous LoF. Nine of the ten rare high-confidence pLoF variants have a resolved origin (5 maternal, 3 paternal, one carried by both parents) and none is a candidate *de novo* variant. The one homozygous rare pLoF, a 45-bp deletion of the canonical exon 2 donor of *PEX5*, would imply Zellweger spectrum disorder if it were a loss of function. Both parents are homozygous for it, which already argues against rarity (ClinVar classifications of the deletion are conflicting), and the exon has an alternative donor 45 bp downstream that keeps the reading frame. Absence from gnomAD here reflects the length of the deletion, not its frequency. The case is a useful reminder that transcript-level LoF calls need biological context; in this analysis the context comes from the ad hoc tier.

5. What this genome cannot tell us. Interval coverage of MANE coding sequence by the benchmark regions shows that 83 ClinGen disease genes [11] are less than 90% assessable in HG002 v4.2.1, 19 less than half. They include genes in segmental duplications and falsely duplicated regions of GRCh38 (*CBS*, *ICOSLG*, *PRODH*; *CBS*, classic homocystinuria, is a known example [1]), *FXN* (Friedreich ataxia) and GC-rich genes such as *ETFB*, *GNPTG* and *FAM20C*. Finding no pathogenic variant in these genes is not evidence of absence.

Why this belongs in the vepyr paper. The analysis is a single Python session: 4.2 min of annotation of the whole genome with VEP-identical core fields and four plugins, then 1.3 min for everything else, including 57 s to read 12 remote sources (25.9 million records) without downloading, indexing or converting them. Three kinds of annotation with three update rates—Ensembl releases, ClinVar weeks, per-question data—are handled by three mechanisms: a core cache, plugin caches and DataFrame operations on data read from URLs. The biology is not new, which is the point: a known result (depletion of LoF from constrained genes), a known founder allele and a known reference artefact are recovered quickly, reproducibly and in one place, which is what users need when the question, not the method, is new.

2 Main text: Results fragment

→ **MAIN TEXT** Results, new subsection after "Performance tests" (or as the "Annotation with analytical workflows" item of the Results outline). About 290 words; can be cut to the first three sentences and the F11 sentence.

Three tiers of annotation in a whole genome

To show how annotation that changes at different rates is combined in one analysis, we annotated the GIAB HG002 benchmark call set (4,096,123 variants, chromosomes 1–22) with the Ensembl 116 core cache and four plugin caches (ClinVar, SpliceAI, AlphaMissense and CADD) in 4.2 min on a laptop, and joined the resulting frame with 12 sources that were read directly from their URLs by polars-bio and Polars: gnomAD gene constraint, ClinGen gene–disease validity, Ensembl gene models, GIAB genome stratifications and benchmark regions, and the benchmark VCFs of both parents (Figure 1D; Supplementary Section X). Of 158 variants with a predicted loss-of-function (pLoF) consequence in a MANE Select transcript, filters drawn from the three tiers—escape from nonsense-mediated decay and near-fixed alleles (core cache), SpliceAI and ClinVar (plugins), and segmental duplications, low mappability and the 50-bp rule (ad hoc sources)—retained 57 high-confidence pLoF variants in 52 genes. pLoF variants were depleted from LoF-intolerant genes (9.5% vs 24.7% of rare synonymous variants in the two lowest LOEUF deciles, $p = 2.9 \times 10^{-4}$), the signature of purifying selection in a single genome. The only pathogenic ClinVar variant with review criteria was the Ashkenazi founder allele *F11* p.Glu135Ter, heterozygous and inherited from the father, with a frequency of 1.85% in Ashkenazi Jewish gnomAD samples. Interval coverage showed that 83 of 2,302 ClinGen genes have less than 90% of their coding sequence within the benchmark regions, including *CBS*, which lies in a falsely duplicated region of GRCh38. The ad hoc sources were read in 57 s, and the analysis after annotation took 1.3 min.

3 Main text: Discussion / Conclusion fragment

→ **MAIN TEXT** Conclusion (`vepyr-main.tex`, "The *vepyr* . . . "; matches the "recurrent vs. ad hoc analysis" note there). About 120 words.

The HG002 example separates annotation by how often it changes rather than by where it comes from. Core annotation follows Ensembl releases and plugin data follow their providers; both are compiled once into indexed Parquet caches, so a weekly ClinVar release rebuilds a 0.1 GiB plugin cache, not the core cache. Data that a particular question needs—gene constraint, curated gene–disease relationships, genome stratifications or the genotypes of relatives—need no cache at all: they are read from their URLs when the query runs and combined with the annotation through joins and interval operations. In a VEP-based workflow, each such source is configured as a custom annotation or plugin before annotation, or processed afterwards from an annotated VCF with separate tools. Keeping the third tier outside the annotation engine lets analyses change without changing, or re-running, the annotation.

4 Main text: Figure 1D

→ **MAIN TEXT** Figure 1, panel D, replacing the chromosome 22 "Beyond `filter_vep`" panel; generated by `paper/figure1/build_tiered_annotation.py`.

The panel below replaces the current panel D ("Beyond `filter_vep`", chromosome 22, segmental duplications) of the cache-workflow version of Figure 1. It is generated by `paper/figure1/build_tiered_annotation.py` in the same draw.io/SVG style as the other panels and reads its numbers from `results/`. The chromosome 22 segmental-duplication example can stay in the supplement as the bedtools-verified equivalence check.

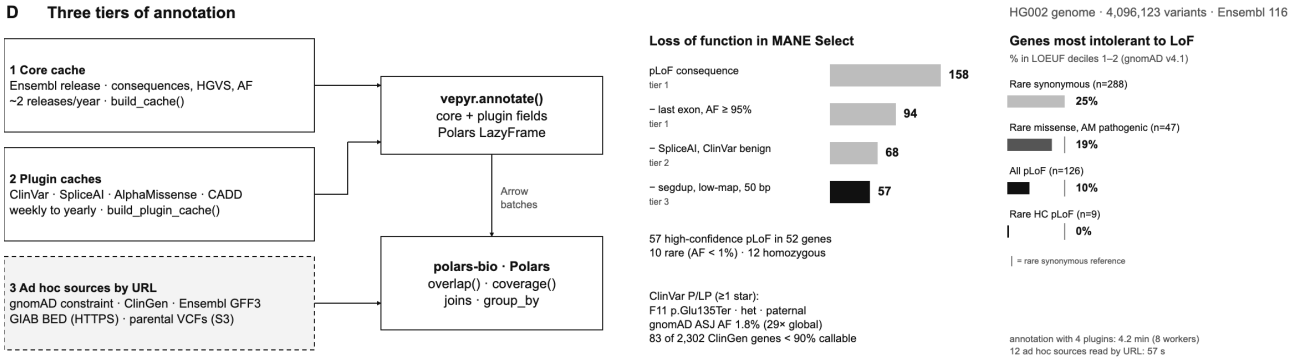


Figure 1D (proposal). Three tiers of annotation in the HG002 genome. Core (tier 1) and plugin (tier 2) annotations are produced by `vepyr.annotate()` from prebuilt Parquet caches that are refreshed at different rates; ad hoc sources (tier 3) are read directly from their URLs by polars-bio and Polars when the query runs and are combined with the annotated frame by joins and interval operations (`overlap()`, `coverage()`). Middle: variants with a predicted loss-of-function (pLoF) consequence in a MANE Select transcript that remain after the filters contributed by each tier (last exon; SpliceAI support and ClinVar benign classification; segmental duplications, low mappability and the 50-bp rule), leaving 57 high-confidence pLoF variants in 52 genes. Right: share of each variant class in the 20% of genes most intolerant to LoF (gnomAD v4.1 LOEUF deciles 1–2); ticks mark the share of rare synonymous variants. pLoF variants are depleted from these genes, as expected under purifying selection. Bottom of the middle column: the only ClinVar pathogenic variant with review criteria, the Ashkenazi founder allele *F11* p.Glu135Ter, with its parental origin from the parental VCFs, and the number of ClinGen genes with less than 90% of their coding sequence in the benchmark regions. Details in Supplementary Section X.

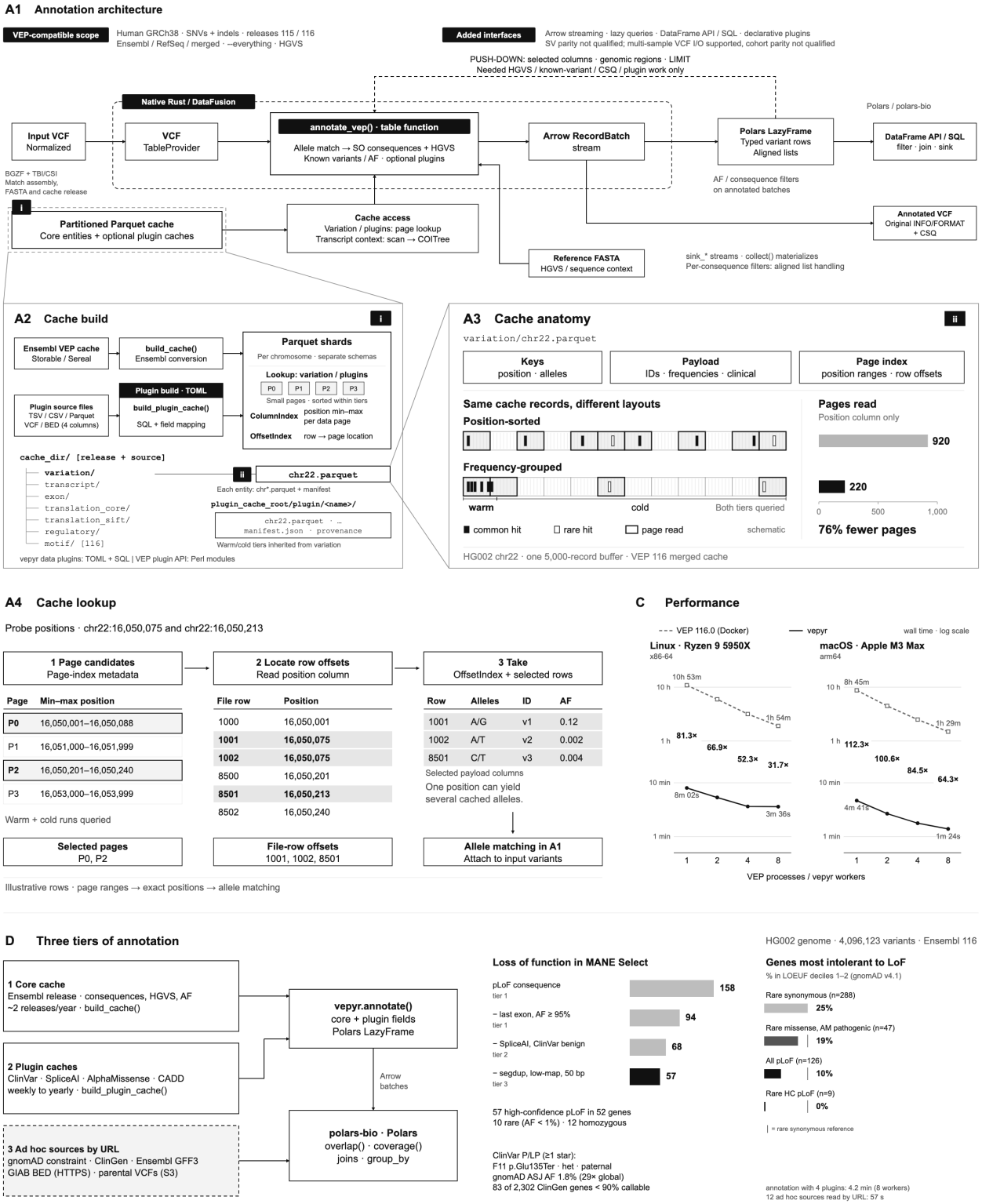


Figure 1 (cache-workflow proposal) with the new panel D in place of the previous one, for checking proportions. Panels A1–A4 and C are unchanged.

5 Supplementary section (condensed, recommended)

→ **SUPPLEMENTARY** New section of `vepyr-supplementary.tex`, after "Beyond `filter_vep`: interval operations". About two pages: the three tiers, one listing, one figure, the biological and performance results. Appendix A is the full version from which it is condensed.

Three tiers of annotation in a whole genome

Annotation sources change at different rates, and vepyr handles them with three mechanisms (Table 1): the core cache, which follows Ensembl releases; plugin caches, which follow their providers; and ad hoc sources, which a particular analysis needs and which are read from their URLs when the query runs, without a build step. We applied all three to the GIAB HG002 benchmark call set (NIST v4.2.1, chromosomes 1–22, 4,096,123 variants after decomposition) to ask which genes a healthy individual inactivates, whether selection against such variants is visible in a single genome, which clinically classified variants HG002 carries and from which parent, and in which disease genes a negative result would be uninformative.

Tier	Data used here	Updated	Mechanism
1 Core cache	Ensembl 116: consequences, HGVS, MANE, gnomAD frequencies (incl. Ashkenazi Jewish)	with Ensembl releases	`build_cache()`; read by `annotate()`
2 Plugin caches	ClinVar (2026-07-06), SpliceAI, AlphaMissense, CADD v1.7	weekly to yearly	`build_plugin_cache()`; matched by `annotate()`
3 Ad hoc sources	gnomAD v4.1 gene constraint; ClinGen gene–disease validity; Ensembl 116 GFF3; GIAB v3.6 stratifications; v4.2.1 benchmark regions and VCFs of HG002 and both parents (12 files, 25.9 million records)	per analysis	read by URL with polars-bio (BED over HTTPS, VCF from S3) or Polars; joins, `overlap()`, `coverage()`

Table 1: Tiers of annotation in the HG002 analysis. Tiers 1 and 2 took 255 s for 4,096,123 variants (8 workers, Apple M3 Max); reading all tier-3 sources took 57 s.

The analysis is a single Python script; the listing shows its structure (abridged; the complete code is in `paper/hg002-tiered-annotation`):

```
lf = vepyr.annotate("HG002_v4.2.1.split.vcf.gz", "vepyr_cache/116_GRCh38_ensembl",
                    reference_fasta=FASTA, workers=8,                    # tier 1
                    plugin_cache_root="vepyr_plugin_cache",              # tier 2
                    plugins=["clinvar", "alphamissense", "spliceai", "cadd"])
lf.select(*COLUMNS).sink_parquet("HG002.parquet")
mane = (pl.scan_parquet("HG002.parquet").with_row_index("vid")
          .explode(CSQ_LIST_COLUMNS, empty_as_null=True)
          .filter(pl.col("MANE_SELECT").is_not_null()).collect())
plof = mane.filter(pl.col("Consequence").str.contains(
    "stop_gained|frameshift_variant|splice_acceptor_variant|splice_donor_variant"))
# tier 3: read from the URLs when the query runs
segdups = pb.scan_bed(f"{GIAB_STRAT}/SegmentalDuplications/GRCh38_segdups.bed.gz")
benchmark = pb.scan_bed(HG002_BENCHMARK_BED)                  # HTTPS
father = pb.scan_vcf(f"s3://giab/release/AshkenazimTrio/{HG003_VCF}",
                     info_fields=[], format_fields=["GT"])
constraint = pl.scan_csv(GNOMAD_V41_CONSTRAINT_TSV, separator="\t", infer_schema=False)
in_segdup = pb.overlap(plof, segdups, overlap_output="left", distinct_output=True)
callable_bp = pb.coverage(mane_cds, benchmark)          # per CDS interval
```

Loss of function. Of 158 variants with a stop-gained, frameshift or canonical splice-site consequence in a MANE Select transcript, 57 in 52 genes pass filters drawn from all three tiers (Figure 1a). Tier 1

removes premature stops in the last exon, which escape nonsense-mediated decay, and alleles with a gnomAD frequency of at least 95%: the *FMO2*2* stop that inactivates *FMO2* in most humans and pairs of frameshifts in *SON* and *SLAIN1* that restore the reading frame where GRCh38 carries a rare sequence. Tier 2 removes variants classified benign in ClinVar, such as *LPL* p.Ser474Ter (S447X), and splice-site variants without SpliceAI support. Tier 3 removes variants in segmental duplications or low-mappability regions, such as the *CYP2D6*4* splice variant, and premature stops covered by the 50-bp rule. pLoF variants are depleted from genes intolerant to LoF: 9.5% lie in the two lowest gnomAD LOEUF deciles, against 24.7% of rare synonymous variants ($p = 2.9 \times 10^{-4}$; Figure 1b), the signature of purifying selection in one genome.

Clinically classified variants and inheritance. The only ClinVar pathogenic or likely pathogenic variant with review criteria is *F11* p.Glu135Ter (Glu117Ter in legacy numbering), the Ashkenazi founder allele of factor XI deficiency, heterozygous in HG002. Its gnomAD frequency is 1.85% in Ashkenazi Jewish samples and 0.063% overall, and the father's VCF, streamed from S3, shows that it was inherited from him. The parental VCFs resolve the origin of 9 of the 10 rare high-confidence pLoF variants and show that the only homozygous one, a 45-bp deletion of the *PEX5* exon 2 donor that is absent from gnomAD, is homozygous in both parents; with an alternative in-frame donor 45 bp downstream, it is most probably not a loss of function.

What cannot be seen. `coverage()` of MANE Select coding sequence by the benchmark regions shows that 83 of 2,302 ClinGen disease genes have less than 90% of their coding sequence assessable (Figure 1c), mostly in GIAB difficult regions; they include *CBS*, which lies in a falsely duplicated region of GRCh38. A negative result in these genes is not informative.

Performance. Annotation of the whole genome with the core cache and four plugin caches took 4.2 min; everything else, including reading the twelve ad hoc sources, took 1.3 min, in one Python process and without intermediate VCF, BED or index files. Changing a tier-3 question—another gene set, stratification or relative—changes a URL, not a cache.

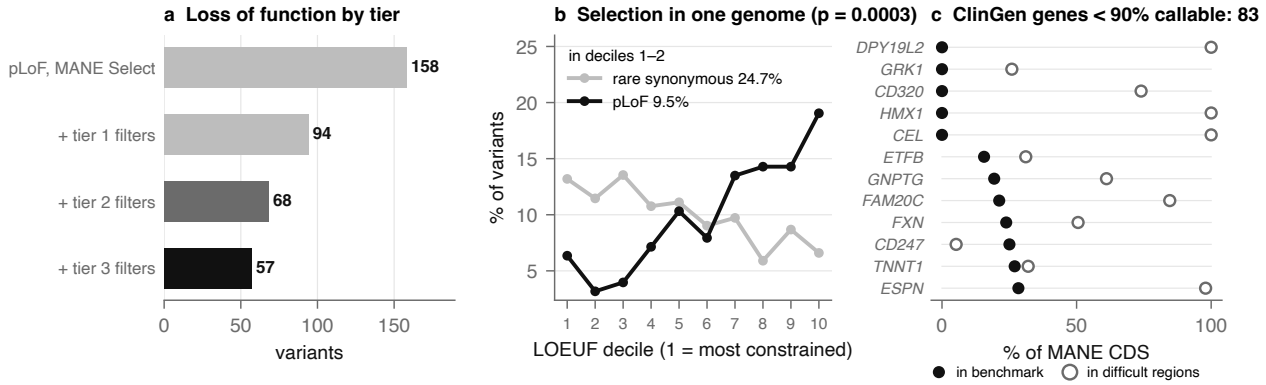


Figure 1: Three tiers of annotation in the HG002 genome. (a) Variants with a MANE Select pLoF consequence remaining after the filters of each tier. (b) Distribution of pLoF and rare synonymous variants across gnomAD v4.1 LOEUF deciles; the legend gives the share in the two most constrained deciles (p, Fisher's exact test). (c) The twelve ClinGen genes with the smallest share of MANE Select coding sequence in the HG002 benchmark regions, and the share in GIAB difficult regions.

6 Notes on the analysis

NOT FOR THE PAPER Observations made while preparing the analysis, including issues in polars-bio that affect claims about remote data.

1. **polars-bio and HTTP(S) sources.** With polars-bio 0.36.2, `scan_bed()` reads BED files over HTTPS, but `scan_vcf()` and `scan_gff()` abort on an `https://` path with "`PanicException: not implemented: Unsupported storage type: HTTP`" (`bio-format-vcf/src/physical_exec.rs:2719`, `bio-format-gff/src/physical_exec.rs:1244`); the VCF and GFF dispatchers handle only local files and GCS/S3/Azure object stores. The analysis therefore reads the parental VCFs from the GIAB mirror on S3 (`s3://giab/...`, anonymous access) and the Ensembl GFF3 with `polars.read_csv()`. If the paper states that polars-bio reads remote genomic files "by URL", HTTP(S) support for VCF and GFF in datafusion-bio-formats should land first; the panic should in any case become a Python exception.

2. **Region pushdown on remote VCF.** A `chrom`/`start` filter on the S3 VCF (indexed with `.tbi`) took as long as a full scan (about 110 s for 4.0 million records from Warsaw to `us-east-1`), i.e. the index is not used for remote files. The analysis streams each parental VCF once and joins it to the candidate variants instead.

3. **Live sources.** The ClinGen gene–disease validity file is regenerated daily (the copy used here was read on October 15, 2026); for the published version, record its creation date and checksum, or pin a snapshot. gnomAD v4.1, GIAB v3.6/v4.2.1 and Ensembl 116 are versioned releases.

4. **Scope of the LoF filters.** The filters are modelled on LOFTEE (last exon, 50-bp rule, splice-site support) but are not LOFTEE: ancestral alleles, GERP, non-canonical splice motifs and the LOFTEE flags are not reproduced, and the published SpliceAI cache holds masked SNV scores only, so splice-site indels are not checked by SpliceAI. The wording in the paper should remain "LOFTEE-inspired" or name the filters explicitly.

5. **Benchmark regions.** The GIAB v4.2.1 benchmark covers chromosomes 1–22 and excludes many difficult regions, so all counts are restricted to the benchmark and are lower bounds for the whole genome; the callability analysis quantifies this per gene. The newer HG002 Q100/T2T-based benchmark would extend coverage, but would need a GRCh38 liftover policy.

6. **Reporting findings in a named genome.** HG002 (PGP participant, open consent) is used in many publications; findings are reported here as ClinVar assertions and population frequencies only, without clinical interpretation. A sentence to that effect may be appropriate in the supplement.

7. **Environment.** The analysis used a clean environment with the PyPI releases vepyr 0.10.0, Polars 1.44.2 and polars-bio 0.36.2; the repository development `.venv` on the authoring machine still has polars-bio 0.26.0 and a stale native module (`import vepyr` fails there).

8. **References.** The bibliography of this document was written from memory for the draft; verify each entry (volume, pages, year) before adding the keys to `references.bib`.

9. **Haplotype-aware consequences.** The compensating frameshift pairs in *SON* and *SLAIN1* are annotated as two frameshifts because VEP (and vepyr) annotate each allele separately; the frequency filter catches them here, but a haplotype-aware consequence mode would be the principled fix.

A Extended supplementary section

→ SUPPLEMENTARY (optional) Full version with all methods, listings, figures and tables. Use instead of Section 5 if space allows, or keep for reviewers; individual subsections (e.g. the rare pLoF table or the callability table) can also be added to the condensed section.

Annotation sources change at different rates. The Ensembl core annotation (transcript models, consequences, HGVS, known variants and population frequencies) changes with Ensembl releases, about twice a year; resources such as ClinVar change weekly, and prediction scores (SpliceAI, AlphaMissense, CADD) with new model releases; and many questions need data that are not annotation sources at all: gene-level constraint, curated gene–disease relationships, gene models, genome stratifications or the genotypes of relatives. vepyr maps the first two onto prebuilt Parquet caches (tiers 1 and 2 below) and leaves the third to the DataFrame ecosystem: such sources are read when a query needs them, directly from their URLs, without a build step (Table 2).

Tier	Sources used here	Changes	Mechanism
1 Core cache	Ensembl 116 `ensembl` cache: consequences, HGVS, MANE, gnomAD frequencies	with Ensembl releases (about twice a year)	`build_cache()`; Parquet; read during annotation
2 Plugin caches	ClinVar (2026-07-06), SpliceAI, AlphaMissense, CADD v1.7	weekly (ClinVar) to yearly	`build_plugin_cache()` from TOML manifests; Parquet; matched during annotation
3 Ad hoc sources	gnomAD v4.1 constraint, ClinGen gene validity, Ensembl 116 GFF3, GIAB v3.6 stratifications, GIAB v4.2.1 benchmark regions and parental VCFs	per analysis	read from the URL by polars-bio or Polars when the query runs; joins and interval operations

Table 2: The three tiers of annotation used in this section.

The example combines the three tiers in one analysis of the GIAB HG002 genome: which predicted loss-of-function (pLoF) variants does a healthy individual carry, does purifying selection against them show in a single genome, which clinically classified variants does HG002 carry and from which parent, and in which clinically relevant genes would a negative result be uninformative because the gene lies outside the benchmark regions?

A.1 Data and software

The input is the NIST v4.2.1 GRCh38 benchmark call set of HG002 (chromosomes 1–22), with multi-allelic records decomposed as in the quick start, which yields 4,096,123 records. The analysis used the `ensembl` core cache of release 116, the ClinVar, AlphaMissense, SpliceAI and CADD plugin caches (chromosomes 1–22 only: 30 GiB and 90 GiB on disk), the Ensembl primary-assembly FASTA, vepyr 0.10.0, Polars 1.44.2 and polars-bio 0.36.2 on an Apple MacBook Pro (M3 Max, 64 GiB). The ad hoc sources are listed in Table 3; none is stored locally.

```
python -m venv hg002 && . hg002/bin/activate
pip install "vepyr[polars-bio]==0.10.0" scipy matplotlib
CHR=$(for c in $(seq 1 22); do printf -- "--chrom %s " $c; done)
vepyr download-cache --release 116 --cache_type ensembl --dir_cache vepyr_cache $CHR
for p in clinvar alphamissense spliceai cadd; do
  vepyr download-plugin-cache --plugin $p --release 116 \
    --plugin_cache_root vepyr_plugin_cache $CHR
done
bcftools norm -m -both -Oz -o HG002_v4.2.1.split.vcf.gz \
  HG002_GRCh38_1_22_v4.2.1_benchmark.vcf.gz
tabix -p vcf HG002_v4.2.1.split.vcf.gz      # 4096123 records
```

A.2 Tiers 1 and 2: annotation

Core and plugin fields are produced by a single call. The annotated frame, restricted to the 33 columns used later (including the gnomAD Ashkenazi Jewish frequencies `gnomADe_ASJ_AF` and `gnomADg_ASJ_AF`, since HG002 is of Ashkenazi Jewish ancestry), is written once to Parquet; the per-consequence fields remain aligned lists:

```
import polars as pl
import polars_bio as pb
import vepyr

lf = vepyr.annotate(
    "HG002_v4.2.1.split.vcf.gz",
    "vepyr_cache/116_GRCh38_ensembl",
    reference_fasta="Homo_sapiens.GRCh38.dna.primary_assembly.fa",
    workers=8,
    plugin_cache_root="vepyr_plugin_cache",
    plugins=["clinvar", "alphamissense", "spliceai", "cadd"],
)
lf.select(*VARIANT_COLS, *CSQ_COLS).sink_parquet("HG002.annotated.parquet")
```

Annotation of the 4,096,123 records with the four plugins took 255 s (4.2 min) with 8 workers.

A.3 Tier 3: ad hoc sources read from their URLs

Twelve sources are read (Table 3). Interval sources are read with polars-bio, which converts BED to 1-based closed coordinates, matching the vepyr frame; the parental VCFs are streamed from the GIAB mirror on Amazon S3. Tabular gene-level sources and the GFF3 gene models are read with Polars:

```
GIAB = "https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/release"
STRAT = f"{GIAB}/genome-stratifications/v3.6/GRCh38@all"
segdups = pb.scan_bed(f"{STRAT}/SegmentalDuplications/GRCh38_segdups.bed.gz")
benchmark = pb.scan_bed(f"{GIAB}/AshkenazimTrio/HG002_NA24385_son/NISTv4.2.1/"
                        "GRCh38/HG002_GRCh38_1_22_v4.2.1_benchmark_noinconsistent.bed")
father = pb.scan_vcf("s3://giab/release/AshkenazimTrio/HG003_NA24149_father/"
                     "NISTv4.2.1/GRCh38/HG003_GRCh38_1_22_v4.2.1_benchmark.vcf.gz",
                     info_fields=[], format_fields=["GT"])
constraint = pl.scan_csv("https://storage.googleapis.com/gcp-public-data--gnomad/"
                         "release/4.1/constraint/gnomad.v4.1.constraint_metrics.tsv",
                         separator="\t", infer_schema=False)
clingen = pl.read_csv("https://search.clinicalgenome.org/kb/gene-validity/download",
                      skip_rows=4, infer_schema=False)
```

A.4 Definition of loss-of-function variants

The frame is expanded to one row per variant and MANE Select transcript (1,787,380 rows). A pLoF is a stop-gained, frameshift, splice-acceptor or splice-donor consequence, the four classes used by LOFTEE. Filters modelled on LOFTEE are then taken from each tier (Table 4): from the core annotation, premature stops in the last exon, which escape nonsense-mediated decay, and alleles with a gnomAD frequency of at least 95%, for which the reference genome carries the rare allele and the "loss" is shared by nearly all humans; from the plugins, canonical splice-site variants with all four SpliceAI delta scores below 0.2 and variants classified benign in ClinVar; and from the ad hoc sources, premature stops within the last 50 bp of the penultimate exon (the 50-bp rule, using exon coordinates from the Ensembl GFF3) and variants in segmental duplications or low-mappability regions, found with `overlap()`:

```
rec = (pl.scan_parquet("HG002.annotated.parquet").with_row_index("vid")
         .explode(CSQ_LIST_COLS, empty_as_null=True)
```

Source	Reader	Records	Time (s)
GIAB v3.6 segmental duplications (BED)	polars-bio scan_bed (HTTPS)	7,432	1
GIAB v3.6 low mappability, all (BED)	polars-bio scan_bed (HTTPS)	673,815	2
GIAB v3.6 all difficult regions (BED)	polars-bio scan_bed (HTTPS)	4,841,150	6
GIAB v3.6 GRCh38 false duplications (BED)	polars-bio scan_bed (HTTPS)	12	0
HG002 v4.2.1 benchmark regions (BED)	polars-bio scan_bed (HTTPS)	481,622	2
HG003 v4.2.1 benchmark regions (BED)	polars-bio scan_bed (HTTPS)	554,108	3
HG004 v4.2.1 benchmark regions (BED)	polars-bio scan_bed (HTTPS)	548,306	2
gnomAD v4.1 gene constraint (TSV)	Polars scan_csv (HTTPS)	17,481	3
ClinGen gene–disease validity (CSV)	Polars read_csv (HTTPS)	3,705	2
Ensembl 116 GFF3 (gene models)	Polars read_csv (HTTPS)	10,723,833	15
HG003 (father) v4.2.1 benchmark VCF	polars-bio scan_vcf (S3)	4,000,097	11
HG004 (mother) v4.2.1 benchmark VCF	polars-bio scan_vcf (S3)	4,031,346	9

Table 3: Ad hoc (tier 3) sources, read from their URLs on every run of the analysis. Times are wall-clock times of a single read from Warsaw and include transfer; the full URLs are in `analysis.py`.

```
        .filter(pl.col("MANE_SELECT").is_not_null())
        .collect())
plof = rec.filter(pl.col("Consequence").str.contains(
    "stop_gained|frameshift_variant|splice_acceptor_variant|splice_donor_variant"))
in_segdup = pb.overlap(plof.select("chrom", "start", "end", "vid"),
                       segdups, overlap_output="left", distinct_output=True)
```

A variant is a high-confidence (HC) pLoF if at least one of its MANE Select pLoF consequences passes all filters. Allele frequencies are the larger of the gnomAD exome and genome frequencies; "rare" denotes a frequency below 1% (or absence from gnomAD).

A.5 Loss-of-function variants in HG002

Of the 4,096,123 variants, 20,725 have a coding or canonical splice-site consequence in a MANE Select transcript and 158 a pLoF consequence. The filters retain 57 high-confidence pLoF variants in 52 genes (Figure 2); 10 are rare, 47 have a frequency of at least 1%, and 12 are homozygous. Each tier removes a different kind of false or uninformative LoF call (Table 4). The frequency filter of tier 1 removes pairs of nearby frameshifts at a frequency of 100% that together restore the reading frame, in *SON* (+1 and −1 nt) and *SLAIN1* (+2 and +1 nt), i.e. positions at which GRCh38 carries a rare sequence, and the *FMO2* p.Gln472Ter allele (*FMO2*2*, 99.5%), which inactivates *FMO2* in most humans. ClinVar removes, among others, the *LPL* p.Ser474Ter allele (S447X in the legacy numbering), a stop-gain that is classified benign, and the *CYP2D6*4* splice-acceptor variant, which is classified as a drug-response allele; the latter also lies in a segmental duplication, as do pLoF calls in olfactory receptor, *NBPF*, *GOLGA8* and *NPIPB* genes.

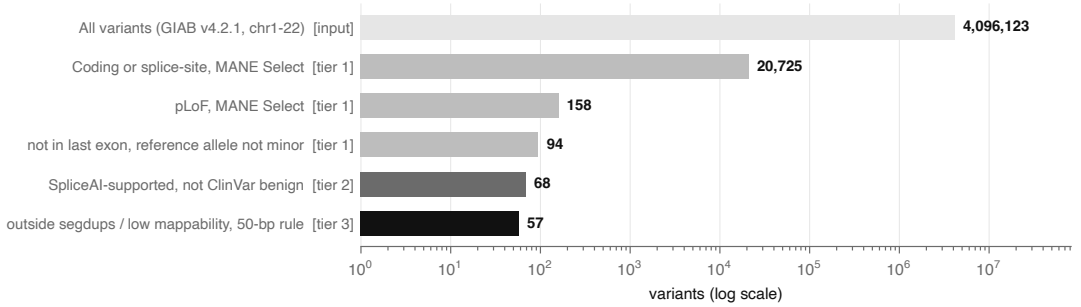


Figure 2: Variants of HG002 retained at each step. The pLoF steps count variants with at least one MANE Select pLoF consequence that passes the filters of the tiers applied so far (log scale; shading by tier).

Tier	Flag	Source	Variants
1	last exon	`EXON` of the MANE Select consequence	55
1	allele frequency $\geq 95\%$	gnomAD exomes/genomes in the core cache	13
2	no SpliceAI support	SpliceAI plugin, all delta scores < 0.2	3
2	ClinVar benign	ClinVar plugin, (likely) benign, not pathogenic	39
3	50-bp rule	Ensembl 116 GFF3 exon coordinates	4
3	segmental duplication	GIAB v3.6 BED, `overlap()`	24
3	low mappability	GIAB v3.6 BED, `overlap()`	10

Table 4: Flags applied to the 158 pLoF variants. A variant can carry several flags; 100 carry at least one.

A.6 Selection against loss of function in a single genome

Genes intolerant to LoF were taken from the gnomAD v4.1 LOEUF metric; the two lowest deciles contain the 20% most constrained genes. Because longer genes, which are more often constrained, carry more variants, rare synonymous variants of HG002 serve as the reference: 24.7% of them fall in the two lowest deciles. pLoF variants are depleted from these genes (9.5%; odds ratio 0.32, Fisher's exact test $p = 2.9 \times 10^{-4}$), and their share generally increases towards the most tolerant genes (Figure 3). None of the 9 rare high-confidence pLoF variants with a LOEUF value lies in the two lowest deciles. Rare missense variants are depleted to a lesser extent (15.7% for those AlphaMissense predicts as likely benign, 19.1% for likely pathogenic; Table 5).

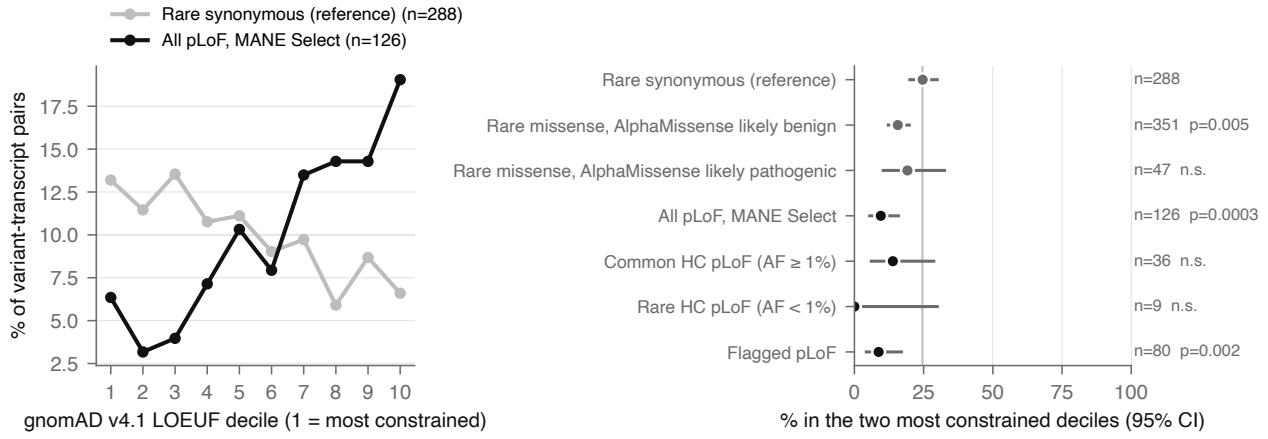


Figure 3: Variant classes of HG002 across gnomAD v4.1 LOEUF deciles. Left: distribution of all MANE Select pLoF variants and of rare synonymous variants. Right: share in the two most constrained deciles with 95% Wilson intervals; p-values from Fisher's exact test against rare synonymous variants. Units are variant–transcript pairs in MANE Select transcripts with a LOEUF value.

Variant class	n	Deciles 1–2 (%)	Odds ratio	p	Median LOEUF
Rare synonymous (reference)	288	24.7	–	–	0.88
Rare missense, AM likely benign	351	15.7	0.57	0.005	1.00
Rare missense, AM likely pathogenic	47	19.1	0.72	0.466	1.01
All pLoF, MANE Select	126	9.5	0.32	2.9×10^{-4}	1.22
High-confidence pLoF, AF $\geq 1\%$	36	13.9	0.49	0.210	1.12
High-confidence pLoF, AF $< 1\%$	9	0.0	0.00	0.121	0.95
Flagged pLoF	80	8.8	0.29	0.002	1.24

Table 5: Share of each variant class in the two most LoF-intolerant LOEUF deciles. AM, AlphaMissense; odds ratios and Fisher's exact test against rare synonymous variants.

A.7 Clinically classified variants and their parental origin

A single HG002 variant has a pathogenic or likely pathogenic ClinVar classification with at least one review star: *F11* p.Glu135Ter (Table 6), the factor XI "type II" allele (Glu117Ter in the legacy numbering of the mature protein), heterozygous and inherited from the father. Its gnomAD frequency is 0.063% overall but 1.85% in the Ashkenazi Jewish samples, 29-fold higher, consistent with its known founder origin and with the ancestry of the trio. Founder effects affect the rare pLoF variants more generally: 4 of the 10 variants that are rare in gnomAD overall have a frequency of at least 1% in the Ashkenazi Jewish samples. ClinGen curates *F11* as semidominant; the variant would therefore be reported as a carrier finding with possible partial factor XI deficiency, which this analysis does not assess.

Streaming the parental benchmark VCFs from S3 and the parental benchmark regions over HTTPS resolves the origin of 9 of the 10 rare high-confidence pLoF variants (5 maternal, 3 paternal, one homozygous variant carried by both parents); none is a candidate *de novo* variant. The unresolved variant, a 168-bp deletion in *HLA-DQB1*, lies outside the parental benchmark regions. The homozygous variant illustrates a limit of transcript-level LoF calling: a 45-bp deletion that removes the canonical exon 2 splice donor of the MANE Select transcript of *PEX5*, whose biallelic loss of function causes Zellweger spectrum disorder. The same exon has an alternative donor 45 bp downstream, and transcripts using it are annotated with an in-frame deletion. Both parents are homozygous for the deletion, which is absent from gnomAD and has conflicting classifications in ClinVar; the absence from gnomAD reflects the length of the deletion rather than its rarity, and the variant is most probably not a loss of function.

Gene	Position	HGVS	Consequence	GT	AF	ASJ AF	ClinVar	MOI	Origin
ZDHHC6	10:112,445,383	p.Arg18AsnfsTer9	frameshift	het	4.1e-06	2.3e-04			paternal
INSC	11:15,190,800	p.Arg227ProfsTer3	frameshift	het	7.9e-05	0.003			paternal
OR5M3	11:56,470,542	c.-44-6_-44-2del	splice acceptor	het	0.008	0.012			maternal
PEX5	12:7,190,512	c.147+77_147+121del	splice donor	hom	0	–	conflicting	AR	both
PXN	12:120,215,644	p.Gln773LysfsTer61	frameshift	het	6.1e-04	0.023			maternal
PLEKHG3	14:64,737,373	p.Lys468Ter	stop gained	het	6.9e-07	4.0e-05			maternal
EFCAB13	17:47,324,524	c.-248+1G>A	splice donor	het	0.001	0.002			maternal
F11	4:186,274,193	p.Glu135Ter	stop gained	het	6.3e-04	0.018	P (2★)	SD	paternal
HLA-DQB1	6:32,664,663	c.346_379+134del	splice donor	het	0	–			unresolved
MSR1	8:16,155,085	p.Arg293Ter	stop gained	het	0.010	0.014	conflicting		maternal

Table 6: Rare (gnomAD AF $< 1\%$) high-confidence pLoF variants of HG002. AF: maximum of the gnomAD exome and genome frequencies; ASJ AF: Ashkenazi Jewish samples; MOI: ClinGen mode of inheritance (Moderate or stronger gene–disease validity; SD, semidominant); origin from the parental benchmark VCFs.

A.8 Callability of clinically relevant genes

A negative result is informative only where the genome has been assessed. `coverage()` of the MANE Select coding sequence by the HG002 benchmark regions shows that 1,134 of 18,352 genes and 83 of the 2,302 genes with a ClinGen gene–disease relationship of at least moderate strength have less than 90% of their coding sequence in the benchmark (19 less than 50%; Figure 4, Table 7). For 37 of these genes, most of the coding sequence lies in GIAB difficult regions, such as segmental duplications, tandem repeats or extreme GC content. *CBS*, *ICOSLG*, *PRODH* overlap regions that are falsely duplicated in GRCh38; *CBS*, the gene of classic homocystinuria, has 39% of its coding sequence in the benchmark.

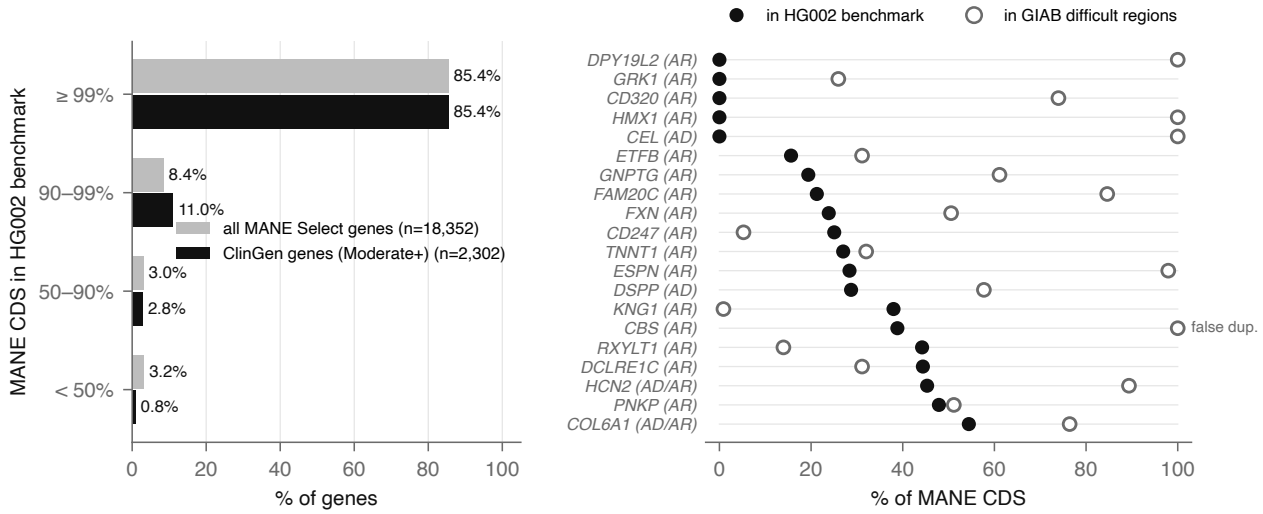


Figure 4: Share of the MANE Select coding sequence within the HG002 v4.2.1 benchmark regions. Left: genes binned by that share. Right: the 20 least callable ClinGen genes, with the share of their coding sequence in GIAB difficult regions (union of all v3.6 stratifications).

Gene	CDS (bp)	Benchmark (%)	Segdup (%)	Difficult (%)	MOI	Disease (ClinGen)
DPY19L2	2,277	0	100	100	AR	male infertility due to globozoospermia
GRK1	1,692	0	0	26	AR	Oguchi disease
CD320	849	0	0	74	AR	methylmalonic acidemia due to...
HMX1	1,047	0	0	100	AR	oculoauricular syndrome
CEL	2,262	0	100	100	AD	maturity-onset diabetes of the young type 8
ETFB	768	16	0	31	AR	multiple acyl-CoA dehydrogenase deficiency
GNPTG	918	19	0	61	AR	GNPTG-mucolipidosis
FAM20C	1,755	21	0	85	AR	lethal osteosclerotic bone dysplasia
FXN	633	24	24	51	AR	Friedreich ataxia
CD247	495	25	0	5	AR	immunodeficiency 25
TNNT1	837	27	0	32	AR	nemaline myopathy 5
ESPN	2,565	28	83	98	AR	nonsyndromic genetic hearing loss
DSPP	3,906	29	0	58	AD	dentinogenesis imperfecta
KNG1	1,935	38	0	1	AR	congenital high-molecular-weight kininogen...
CBS	1,656	39	100	100	AR	classic homocystinuria
RXYLT1	1,332	44	0	14	AR	muscle-eye-brain disease
DCLRE1C	2,079	44	26	31	AR	severe combined immunodeficiency due to...
HCN2	2,670	45	41	89	AD/AR	complex neurodevelopmental disorder
PNKP	1,566	48	0	51	AR	microcephaly, seizures, and developmental...
COL6A1	3,087	54	0	76	AD/AR	collagen 6-related myopathy
ACAN	7,707	59	0	41	AD	ACAN-related short stature spectrum
KANSL1	3,318	61	4	11	AD	Koolen-de Vries syndrome
PRG4	4,215	64	0	39	AR	camptodactyly-arthropathy-coxa...
CASP8	1,440	65	0	1	AR	autoimmune lymphoproliferative syndrome...
COL18A1	4,020	65	0	88	AR	Knobloch syndrome 1

Table 7: The 25 ClinGen genes with the smallest share of MANE Select coding sequence in the HG002 benchmark regions.

A.9 Run time

Annotation of the 4,096,123 variants with the core cache and four plugin caches took 255 s with 8 workers; the remaining analysis, including reading the 12 ad hoc sources (25.9 million records in total) from their URLs in 57 s, took 1.3 min, in a single Python process without intermediate VCF, BED or index files. The tier-3 sources were not converted into caches: a change of the question (another gene set, another stratification, another relative) changes only the URL.

A.10 Conclusions

The example uses each tier for what it does best. The core cache supplies VEP-identical consequences and population frequencies for 4.1 million variants in minutes; plugin caches add frequently updated or licensed evidence without rebuilding the core; and ad hoc sources answer analysis-specific questions—gene constraint, curated disease genes, gene models, genome stratifications and parental genotypes—through joins and interval operations on the annotated frame. Biologically, the single genome reproduces the depletion of LoF variants from constrained genes, recovers the Ashkenazi founder allele *F11* p.Glu135Ter with its parental origin, and identifies the clinically relevant genes in which a negative result would not be informative.

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